Siyuan (Simon) Xing
Curriculum Vitae

July 2026

Department of Mechanical Engineering
California Polytechnic State University, San Luis Obispo, CA 93407
+1 (618) 616-5796
sixing@calpoly.edu

Education

2016–2019	*Southern Illinois University Carbondale*, Illinois, USA • Ph.D. in Engineering Science
2013–2016	*Southern Illinois University Edwardsville*, Illinois, USA • M.S. in Mechanical Engineering
2009–2013	*Sichuan University*, Sichuan, China • B.S. in Electrical Engineering

Employment History

09/2025–Present	Associate Professor, *California Polytechnic State University*, California, USA
02/2026–03/2026	Research Specialist (Visiting Appointment), *Arizona State University*, Arizona, USA • Host: Prof. Ying-Cheng Lai
09/2025–03/2026	Visiting Associate Professor, *Stanford University*, California, USA • Host: Prof. Daniel Tartakovsky
09/2019–09/2025	Assistant Professor, *California Polytechnic State University*, California, USA

Research Interests

Nonlinear Dynamics; Bifurcation Theory; Scientific Machine Learning; Structural Vibrations; Multi-Objective Optimization; Control Design; Bio-Inspired Robotics

Awards and Honors

2025	Travel Award, *2nd SIAM Northern and Central California Sectional Conference (NCC25)*, Lawrence Berkeley National Laboratory (LBNL), October 27–28, 2025
2023	Best Face-to-Face Presentation Award, *International Conference on Nonlinear Science and Complexity, Istanbul, Turkey, July 10–15, 2023*
2023	Lockheed Endowed Professorship, *College of Engineering, California Polytechnic State University*
2023	Chrones Endowed Professorship, *Department of Mechanical Engineering, California Polytechnic State University*
2023	Bentley Professor, *Department of Mechanical Engineering, California Polytechnic State University*
2016	Outstanding Teaching Assistant Award, *Southern Illinois University Edwardsville*
2016	Outstanding Graduate Award, *Southern Illinois University Edwardsville*

Books

1. S. Xing and A. C. Luo, *Sequential Bifurcation Trees to Chaos in Nonlinear Time-delay Systems*. Springer, Cham, 2021, DOI: 10.1007/978-3-031-79669-2.

Book Chapters

1. S. Xing and A. C. Luo, "A sequential order of periodic motions in a 1-d, delayed, nonlinear dynamical system," in *Nonlinear Dynamics, Chaos, and Complexity: In Memory of Valentin Afraimovich (1945–2018)* (Understanding Complex Systems), D. Volchenkov, Ed., Understanding Complex Systems. Springer, Cham, 2021, pp. 95–112, DOI: 10.1007/978-981-15-9034-4_7.
2. A. C. Luo and S. Xing, "Bifurcation trees of period-3 motions to chaos in a time-delayed Duffing oscillator," in *Regularity and Stochasticity of Nonlinear Dynamical Systems*, D. Volchenkov and X. Leoncini, Eds. Springer, Cham, 2018, vol. 21, pp. 247–262, DOI: 10.1007/978-3-319-58062-3_10.
3. A. C. Luo and S. Xing, "Time-delay effects on periodic motions in a duffing oscillator," in *Chaotic, Fractional, and Complex Dynamics: New Insights and Perspectives* (Understanding Complex Systems), M. Edelman, E. E. N. Macau, and M. A. F. Sanjuan, Eds., Understanding Complex Systems. Springer, Cham, 2018, pp. 77–100, DOI: 10.1007/978-3-319-68109-2_5.

Refereed Journal Papers **(students with *)**

1. S. Xing, Q. Han*, and E. Charalampidis, "CombOpNet: A neural-network accelerator for SINDy," *Journal of Vibration Testing and System Dynamics*, vol. 9, no. 1, pp. 1–20, 2025, DOI: 10.5890/JVTSD.2025.03.001.
2. W. Yu*, Y. Xu, X. Wang, L. Hong, J. Jiang, and S. Xing, "Sparse separable Gaussian neural network for interpretable learning of stationary solutions of Fokker–Planck–Kolmogorov equations," *Nonlinear Dynamics*, 2025, DOI: 10.1007/s11071-025-11602-5.
3. X. Wang*, S. Xing, J. Jiang, L. Hong, and J.-Q. Sun, "Separable Gaussian neural networks for high-dimensional nonlinear stochastic systems," *Probabilistic Engineering Mechanics*, vol. 76, 103594 (12 pages), 2024, DOI: 10.1016/j.probengmech.2024.103594.
4. S. Xing and E. G. Charalampidis, "Learning traveling solitary waves using separable Gaussian neural networks," *Entropy*, vol. 26, no. 5, 2024, DOI: 10.3390/e26050396.
5. A. Zhao*, S. Xing, X. Wang, and J.-Q. Sun, "Radial basis function neural networks for optimal control with model reduction and transfer learning," *Engineering Applications of Artificial Intelligence*, vol. 136, p. 108 899, 2024.
6. J. Kochavi* and S. Xing, "Mathematical modeling and system identification of a piezo-actuated, cantilever beam with interferometric measurement," *Journal of Vibration Testing and System Dynamics*, vol. 7, no. 4, pp. 447–461, 2023, DOI: 10.5890/JVTSD.2023.12.004.

7. M. M. Lee*, E. G. Charalampidis, S. Xing, C. Chong, and P. Kevrekidis, "Breathers in lattices with alternating strain-hardening and strain-softening interactions," *Physical Review E*, vol. 107, no. 5, p. 054 208, 2023, DOI: 10.1103/PhysRevE.107.054208.
8. S. Xing and A. C. Luo, "Period-1 motions to twin spiral homoclinic orbits in the Rössler system," *Journal of Computational and Nonlinear Dynamics*, vol. 18, no. 8, 081008 (8 pages), 2023, DOI: 10.1115/1.4062201.
9. S. Xing and A. C. Luo, "Spikes adding to infinity on period-1 orbits to chaos in the Rössler system," *International Journal of Bifurcation and Chaos*, vol. 33, no. 13, 2330033 (16 pages), 2023, DOI: 10.1142/S0218127423300331.
10. S. Xing and J. Sun, "Impulse response of an elastic rod with a mass-damper-spring termination," *Journal of Vibration Testing and System Dynamics*, vol. 7, no. 2, pp. 169–186, 2023, DOI: 10.5890/JVTSD.2023.06.005.
11. S. Xing and J. Sun, "Separable Gaussian neural networks: Structure, analysis, and function approximations," *Algorithms*, vol. 16, no. 10, 453 (19 pages), 2023, DOI: 10.3390/a16100453.
12. S. Xing, B. Kwan*, and P. Duan, "Modeling and control of the locomotion of a monopod robot mounted to a vertical slider," *Journal of Vibration Testing and System Dynamics*, vol. 6, no. 4, pp. 413–430, 2022, DOI: 10.5890/JVTSD.2022.12.005.
13. S. Xing and A. C. Luo, "On an origami structure of period-1 motions to homoclinic orbits in the Rössler system," *Chaos: An Interdisciplinary Journal of Nonlinear Science*, vol. 32, no. 12, 123121 (13 pages), 2022, DOI: 10.1063/5.0131970.
14. S. Xing and A. C. Luo, "Periodic cutting motions in a vibration-assisted, regenerative, nonlinear orthogonal turning system," *International Journal of Dynamics and Control*, vol. 10, pp. 1–12, 2022, DOI: 10.1007/s40435-021-00779-3.
15. S. Xing and J. Sun, "Multi-objective optimization of an elastic rod with viscous termination," *Mathematical and Computational Applications*, vol. 27, no. 6, 94 (13 pages), 2022, DOI: 10.3390/mca27060094.
16. S. Xing and A. C. Luo, "Sequential periodic motions in a vibration-assisted, regenerative, nonlinear turning-tool system," *International Journal of Bifurcation and Chaos*, vol. 31, no. 12, p. 2 150 186, 2021, DOI: 10.1142/S0218127421501868.
17. S. Xing and A. C. Luo, "Understanding dynamics of infinite-equilibrium systems via a quadratic nonlinear system," *Journal of Vibration Testing and System Dynamics*, vol. 5, no. 2, pp. 131–147, 2021, DOI: 10.5890/JVTSD.2021.06.003.
18. L. Chen*, S. Xing, and X. Xiong, "On the analytical modeling of a resonant fatigue testing rig," *Journal of Vibration Testing and System Dynamics*, vol. 4, no. 4, pp. 389–399, 2020, DOI: 10.5890/JVTSD.2020.12.007.
19. S. Xing and A. C. Luo, "On period-1 motions to chaos in a 1-dimensional, time-delay, nonlinear system," *International Journal of Dynamics and Control*, vol. 8, pp. 44–50, 2020, DOI: 10.1007/s40435-019-00546-5.
20. S. Xing and A. C. Luo, "Bifurcation trees of period-1 motions in a periodically excited, softening Duffing oscillator with time delay," *International Journal of Dynamics and Control*, vol. 7, no. 3, pp. 842–855, 2019, DOI: 10.1007/s40435-019-00520-1.

21. S. Xing and A. C. Luo, “On a global sequential scenario of bifurcation trees to chaos in a first-order, time-delayed system,” *International Journal of Bifurcation and Chaos*, vol. 29, no. 10, 1950141 (39 pages), 2019, DOI: 10.1142/S0218127419501414.
22. S. Xing and A. C. Luo, “Periodic motions to chaos in a 1-dimensional, time-delay, nonlinear system,” *The European Physical Journal Special Topics*, vol. 228, no. 9, pp. 1747–1765, 2019, DOI: 10.1140/epjst/e2019-800243-y.
23. S. Xing and A. C. Luo, “On possible infinite bifurcation trees of period-3 motions to chaos in a time-delayed, twin-well Duffing oscillator,” *International Journal of Dynamics and Control*, vol. 6, no. 4, pp. 1429–1464, 2018, DOI: 10.1007/s40435-018-0418-y.
24. A. C. Luo and S. Xing, “Bifurcation trees of period-3 motions to chaos in a time-delayed Duffing oscillator,” *Nonlinear Dynamics*, vol. 88, no. 4, pp. 2831–2862, 2017, DOI: 10.1007/s11071-017-3415-3.
25. A. C. Luo and S. Xing, “Time-delay effects on periodic motions in a periodically forced, time-delayed, hardening Duffing oscillator,” *Journal of Vibration Testing and System Dynamics*, vol. 1, no. 1, pp. 73–91, 2017, DOI: 10.5890/JVTSD.2017.03.006.
26. S. Xing and A. C. Luo, “Towards infinite bifurcation trees of period-1 motions to chaos in a time-delayed, twin-well Duffing oscillator,” *Journal of Vibration Testing and System Dynamics*, vol. 1, no. 4, pp. 353–392, 2017, DOI: 10.5890/JVTSD.2017.12.006.
27. A. C. Luo and S. Xing, “Analytical predictions of period-1 motions to chaos in a periodically driven quadratic nonlinear oscillator with a time-delay,” *Mathematical Modeling of Natural Phenomena*, vol. 11, pp. 75–88, 2016, DOI: 10.1051/mmnp/201611206.
28. A. C. Luo and S. Xing, “Multiple bifurcation trees of period-1 motions to chaos in a periodically forced, time-delayed, hardening Duffing oscillator,” *Chaos, Solitons & Fractals*, vol. 89, pp. 405–434, 2016, DOI: 10.1016/j.chaos.2016.02.005.
29. A. C. Luo and S. Xing, “On frequency responses of period-1 motions to chaos in a periodically forced, time-delayed quadratic nonlinear system,” *International Journal of Dynamics and Control*, vol. 5, no. 3, pp. 466–476, 2016, DOI: 10.1007/s40435-015-0222-x.
30. A. C. Luo and S. Xing, “Symmetric and asymmetric period-1 motions in a periodically forced, time-delayed, hardening Duffing oscillator,” *Nonlinear Dynamics*, vol. 85, no. 2, pp. 1141–1166, 2016, DOI: 10.1007/s11071-016-2750-0.

Refereed Conference Papers **(all presented by Siyuan Xing)**

1. S. Xing and J. Sun, “Time-embedding GRBFNN: Learning transient response of stochastic dynamical systems,” in *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Washington, D.C., USA, Aug. 2024, DOI: 10.1115/DETC2024-143365.
2. J. Kochavi and S. Xing, “Mathematical modelling of a piezo-actuated cantilever beam with interferometer feedback,” in *International Mechanical Engineering Congress and Exposition*, Columbus, OH, USA, Nov. 2022, DOI: 10.1115/IMECE2022-95819.
3. S. Xing and A. C. Luo, “Coexisting unstable periodic motions in the Rössler system,” in *International Mechanical Engineering Congress and Exposition*, Columbus, OH, USA, Nov. 2022, DOI: 10.1115/IMECE2022-95826.

4. S. Xing and A. C. Luo, "Period-1 motions to homoclinic orbits in the Rössler system," in *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, St. Louis, MO, USA, Aug. 2022, DOI: 10.1115/DETC2022-91029.
5. S. Xing and A. C. Luo, "Controlling the dynamics of a quadratic oscillator using infinite equilibria," in *International Mechanical Engineering Congress and Exposition*, Online, USA, Aug. 2021, DOI: 10.1115/IMECE2021-71998.
6. J. Huang, X. Fu, Z. Jing, and S. Xing, "Discontinuous dynamics and bifurcation for morphing aircraft switching on the velocity boundary system," in *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Online, USA, Aug. 2020, DOI: 10.1115/DETC2020-22008.
7. S. Xing, A. C. Luo, and J. Huang, "On the dynamics of a quadratic-oscillator-based, infinite-equilibrium system," in *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Online, USA, Aug. 2020, DOI: 10.1115/DETC2020-22233.
8. S. Xing and A. C. Luo, "Period-1 motions in a periodically forced, nonlinear, machine-tool system," in *International Mechanical Engineering Congress and Exposition*, Salt Lake City, UT, USA, Nov. 2019, DOI: 10.1115/IMECE2019-10771.
9. S. Xing and A. C. Luo, "Regenerative cutting dynamics for a periodically forced machine-tool system," in *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Anaheim, CA, USA, Aug. 2019, DOI: 10.1115/DETC2019-97262.
10. S. Xing and A. C. Luo, "Periodic motions in a first-order, time-delayed, nonlinear system," in *International Mechanical Engineering Congress and Exposition*, Pittsburgh, PA, USA, Nov. 2018, DOI: 10.1115/IMECE2018-86824.
11. A. C. Luo and S. Xing, "Analytical prediction of period-1 motions in a time-delayed, softening Duffing oscillator," in *International Mechanical Engineering Congress and Exposition*, Tampa, FL, USA, Nov. 2017, DOI: 10.1115/IMECE2017-70824.
12. A. C. Luo and S. Xing, "Period-3 motions in a periodically forced, damped, double-well Duffing oscillator with time delay," in *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Cleveland, OH, USA, Aug. 2017, DOI: 10.1115/DETC2017-67210.
13. A. C. Luo and S. Xing, "On complex periodic motions in a time-delayed, double-well Duffing oscillator with strong excitation," in *ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, Charlotte, NC, USA, Aug. 2016, DOI: 10.1115/DETC2016-59343.
14. A. C. Luo and S. Xing, "On time-delay effects on period-1 motions in a periodically forced, time-delayed, Duffing oscillator," in *International Mechanical Engineering Congress and Exposition*, Phoenix, AZ, USA, Nov. 2016, DOI: 10.1115/IMECE2016-66198.
15. A. C. Luo and S. Xing, "Bifurcation trees of period-1 motions to chaos in a quadratic nonlinear oscillator with time-delayed displacement," in *International Mechanical Engineering Congress and Exposition*, Houston, TX, USA, Nov. 2015, DOI: 10.1115/IMECE2015-50029.

Presentations and Posters

1. S. Xing, "Separable Gaussian neural networks and their applications in solving fpk equations," 2024, the Online Conference on Days of Applied Nonlinearity and Complexity.
2. S. Xing, "Bifurcation trees of period-1 to period-2 motions in a Rössler system with multiple delays," 2021, Second Online Conference on Nonlinear Dynamics and Complexity, ISEP, Porto, Portugal.
3. S. Xing and A. C. Luo, "On the quantitative analysis of periodic motions in a time-delayed, softening, duffing oscillator," in *4th Annual Meeting of the SIAM Central States Section*, Norman, OK, USA, Oct. 2018.
4. S. Xing and A. C. Luo, "Periodic motion to chaos in a first-order, time-delayed, nonlinear dynamical system," Aug. 2018, Poster presented at *7th International Conference on Nonlinear Science and Complexity*, San Luis Potosí, México.
5. A. C. Luo and S. Xing, "Complete routes of period-1 motions to chaos in a time-delayed duffing oscillator," in *6th International Conference on Nonlinear Science and Complexity*, São José dos Campos, Brazil, May 2016.
6. A. C. Luo and S. Xing, "From period-1 motions to chaos in a time-delayed, quadratic nonlinear oscillator," in *1st Annual Meeting of SIAM Central States Section*, Rolla, MO, USA, Apr. 2015.
7. A. C. Luo and S. Xing, "On bifurcation trees for period-1 motion to chaos in a periodically forced quadratic nonlinear oscillator with time delay," in *6th International Conference on Nonlinear Science and Complexity*, La Manga, Spain, May 2015.

Invited Talks

1. S. Xing, "Data-driven discovery of high-dimensional dynamical systems with sparse interpretable neural networks," Jul. 2026, Invited talk at the International Conference on Nonlinear Science and Complexity (NSC 2026), Thessaloniki, Greece.
2. S. Xing, "On an origami structure of period-1 motions to homoclinic orbits in the Rössler system," Jul. 2023, Invited talk at Hybrid International Conference on Nonlinear Science and Complexity, Istanbul, Turkey.
3. S. Xing, "Biparametric folding-fan structures of periodic motions in the Rössler system," Sep. 2022, Invited talk at 2022 Conference in Nonlinear Dynamics and Complexity, Thessaloniki, Greece.
4. S. Xing, "The implicit mapping method for the prediction of periodic motions in nonlinear systems," Nov. 2021, Invited talk at Shandong Normal University, Shandong, China.
5. S. Xing, "Periodic motions to chaos in a time-delayed, double-well duffing oscillator with strong excitation," Oct. 2019, Invited talk at California Polytechnic State University, San Luis Obispo, CA, USA.
6. A. C. Luo and S. Xing, "An analytical prediction of period-1 motions to chaos in a time-delayed, duffing nonlinear oscillator through implicit mappings," Dec. 2015, Invited talk at Sichuan University of Science & Engineering, Zigong, Sichuan, China.

7. A. C. Luo and S. Xing, "Analytical predictions of period-1 motions to chaos in a periodically driven duffing nonlinear oscillator with a time-delay displacement," Dec. 2015, Invited talk at Southwest Jiaotong University, Chengdu, Sichuan, China.

Funding

In total: $622,550.62; external funding: $429,934; internal funding: $192,616.62.

1. "Lagrangian-Informed Data-Driven Modeling and Control" (PI; Co-PIs: Brendon Anderson and Maria Pantoja), $35,976, *Noyce School, California Polytechnic State University*, 2026/01–2027/06.
2. "Integration of Quanser's Qube Servo into Cal Poly's ME 418 and 419 Control Courses" (PI), $72,527 (equipment donation), *Comcast*, 2024/09.
3. "Lockheed Endowed Professorship: Reinforcement Learning Control for Legged Robots" (PI), $30,000, *College of Engineering, California Polytechnic State University*, 2023/06–2024/06.
4. "Chrones Endowed Professorship" (PI), $50,000, *Department of Mechanical Engineering, California Polytechnic State University*, 2023/07–2025/07.
5. "I. Gait Control of Legged Robots; II. Dynamic Analysis of MEMS Electrostatic Energy Harvesters" (PI), 12 WTU, approximately $30,000, *Donald E. Bently Center for Engineering Innovation, Department of Mechanical Engineering, California Polytechnic State University*, 2023/06–2024/06.
6. "Simulating and Controlling the Locomotion of a Legged Robot in Real Time" (PI), *Cal Poly CENG*, approximately $7,000, 2022/06–2022/09.
7. "Motion Prediction and Control of Legged Robots" (PI), *Donald E. Bently Center for Engineering Innovation, Cal Poly*, 9 WTUs of assigned time, 2022/09–2023/06.
8. "Controller Design for Vibration-Assisted Machining Using Interferometer Feedback" (PI), *Cal Poly CENG R-IDC*, $4,000, 2022/04–2023/06.
9. "8-Degree-of-Freedom Quadruped Senior Project" (PI), *Cal Poly CENG CPConnect*, $3,611, 2022/03–2022/07.
10. "Tune Expert" (PI), *Keysight Technologies, Inc.*, $357,407 (donation: $57,407), 2021/01–2022/06.
11. "Analytical and Experimental Study of a Single-Leg Robot" (PI), *Cal Poly CENG*, $2,700, 2021/03–2022/03.
12. "Validation and Analysis of a Single-Leg Hopping Robot" (Co-PI), *Cal Poly CENG*, $3,329.62, 2021/04–2021/07.
13. "A Roadmap to Energy Harvesting Using Granular Crystal Chains" (Co-PI), RSCA, *Cal Poly*, $18,000, 2020/06–2021/06.
14. "Study of the Locomotion of a Hopping Robot Using Stateflow" (PI), SURP, *Cal Poly CENG*, $8,000 (total costs inclusive of student and faculty stipends), 2020/06–2020/09.

Courses Developed

Assistant Professor	*California Polytechnic State University* Implementation of Mechanical Controls (ME 418)

Courses Taught

Assistant Professor	*California Polytechnic State University* Engineering Dynamics (ME 212) Intermediate Dynamics (ME 326) Mechanical Vibrations (ME 318) Implementation of Mechanical Controls (ME 418) Mechanical Control Systems (ME 422)
Teaching Assistant	*Southern Illinois University Edwardsville* Numerical Simulation (ME 354) Dynamic System Modeling Laboratory (ME 356L) Stress Laboratory (ME 380L)

Teaching Workshops Attended

2022 NETI-1 Course Design and Student Engagement, *National Effective Teaching Institute.*

Master's Thesis Committees

1. Peyton Ulrich, *Design of a Three-phase Brushless DC Motor Control System*, 2021
2. Jashua T. Castle, *Design of a Printed Circuit Board for a Sensorless Three-Phase Brushless DC Motor Control System*, 2020

Advised Students

Graduate Students

1. Jack Butler, M.S., Fall 2023–Present
2. Jeremy Stephen West, M.S., Fall 2023–Present
3. Connor Curtis Getz, M.S., Fall 2023–Spring 2024
   Design and optimization of transmission lines of loudspeakers
4. Andrew Donald Maas, M.S., Winter 2022–Winter 2024
   Modeling and System Identification of an Elastic Beam with a Piezoelectric Patch Exhibiting Hysteresis
5. Scott Dunn, M.S., Winter 2022–Present

6. John Bennett, M.S., Winter 2022–Summer 2024
   Simulation and Real-Time Control of a Single-Legged Robot
7. Tyler Jordan McCue, M.S., 2022–2023
   Reinforcement Learning Control of an 8-DOF Quadruped Robot
8. Jordan Kochavi, M.S., 2021–2022
   System Identification of a Piezo-actuated Cantilever Beam with Interferometer Feedback Using Adaptive Filters
9. Patrick John Ward, M.S., 2021–2022
   Modeling and Control of an 8-DOF Quadruped Robot
10. Craig Kimball, M.S., 2021–2022
    Design of a Proprioceptive Actuator Utilizing a Cycloidal Gearbox
11. Bradley Yuki Nakamoto Kwan, M.S., 2020–2021
    Modeling and Control of a Vertical Hopping Robot

Undergraduate Students

1. Jack Butler, 2022–2023 Spring
2. Baxter James Bartlett, 2022–2023 Spring
3. Phillip Alexander Shafik, 2022–2023 Spring
4. Tarun Sreesaila Ganamur, 2022–2023 Spring
5. Sean Robert Wahl, 2022 Summer
6. Clayton Turner Elwell, 2021–2022
7. Tim Jain, 2021–2022

Community Service

Judge	Senior Design Expo, Department of Mechanical Engineering, 2023
Judge	Senior Design Expo, Department of Mechanical Engineering, 2022
Judge	NSF Student Poster Competition, *International Mechanical Engineering Congress & Exposition*, 2019
Course Coordinator	ME 418: Implementation of Mechanical Controls, since Fall 2022
Lab Coordinator	Control Courses (ME 418 and ME 419), since 2026
Member	Events Committee, *Department of Mechanical Engineering, California Polytechnic State University*, 2024
Member	Hiring Committee, *Department of Mechanical Engineering, California Polytechnic State University*, 2023
Member	Research Support Task Force, *Department of Mechanical Engineering, California Polytechnic State University*, since 2023
Member	Computing Committee, *Department of Mechanical Engineering, California Polytechnic State University*, since 2021
Member	Advancement of Scholarship Task Force, *Department of Mechanical Engineering, California Polytechnic State University*, 2019

Professional Service

- Editorship
  - Associate Editor, *Discontinuity, Nonlinearity, and Complexity*, since 2021.
  - Associate Editor, *Journal of Environmental Accounting and Management*, since 2022.
  - Young Editorial Board Member, *International Journal of Dynamics and Control*, 2022–2025.
- Technical Program Committees
  - Technical Program Committee Member, *3rd Annual SIAM Northern and Central California Section Conference (NCC26)*, Davis, California, USA, October 26–27, 2026.
  - Scientific Committee Member, *International Conference on Nonlinear Science and Complexity (NSC 2026)*, Thessaloniki, Greece, July 14–17, 2026.
  - Member of the International Technical Committee and Steering Committee, *International Conference on Nonlinear Science and Complexity (NSC 2025)*, Rio Claro, Brazil, August 4–8, 2025.
  - *Days of Applied Nonlinearity and Complexity*, online conference hosted by the Physics Department, Aristotle University of Thessaloniki, Greece, January 12–14, 2024.
  - *Hybrid Conference on Nonlinear Science and Complexity*, Istanbul, Turkey, July 10–15, 2023.

- *Online Conference in Nonlinear Dynamics and Complexity*, Greece, September 26–29, 2022.
- *The 2nd Online Conference on Nonlinear Dynamics and Complexity*, online, Central Time Zone, October 4–6, 2021.
- *The 1st Online Conference on Nonlinear Dynamics and Complexity*, online, Central Time Zone, November 23–25, 2020.

➤ Organization of Conferences

- Co-Chair of the online minisymposium "Data-Driven Dynamical Modeling of Complex Mechanical and Vibrating Systems," *The Fifth International Nonlinear Dynamics Conference (NODYCON 2026)*, Rome, Italy, September 20–23, 2026.
- Organizing Committee Member, *2nd Days of Applied Nonlinearity and Complexity (DANOC)*, Aristotle University of Thessaloniki, Greece, January 23–25, 2026.
- Conference Program Chair, *Nonlinear Science and Complexity*, Yibin, Sichuan, China, August 5–10, 2024.
- Chair of the symposium "Nonlinear Dynamics of Engineering Systems," *Hybrid Conference on Nonlinear Science and Complexity*, Istanbul, Turkey, July 10–15, 2023.
- Co-Chair of the minisymposium "Nonlinear Dynamics in Neural and High-Dimensional Systems," *2022 Conference in Nonlinear Dynamics and Complexity*, Greece, September 26–29, 2022.
- Chair of the minisymposium "Nonlinear Vibrations and Waves," *The 2nd Online Conference on Nonlinear Dynamics and Complexity*, Portugal, October 4–6, 2021 (Greenwich Mean Time).
- Co-Organizer of the minisymposium "Vibration and Time-Delay Systems," *The 1st Online Conference on Nonlinear Dynamics and Complexity*, online, Central Time Zone, November 23–25, 2020.
- Conference Co-Organizer, *The 1st Online Conference on Nonlinear Dynamics and Complexity*, online, Central Time Zone, November 23–25, 2020.
- Co-Organizer of the session "Time-Delay Systems and Discontinuous Dynamical Systems," *International Design Engineering Technical Conferences & Computers and Information in Engineering Conference*, Anaheim, California, USA, August 18–21, 2019.

➤ Referee for Scientific Journals and Books

- *Physica D: Nonlinear Phenomena*, since 2026
- *Physica A: Statistical Mechanics and Its Applications*, since 2026
- *Meccanica*, since 2025
- *Journal of Nonlinear Science*, since 2025
- *Journal of Applied Analysis and Computation*, since 2025
- *Journal of Computational Physics*, since 2024

- *Nonlinear Dynamics*, since 2023
- *Journal of Sound and Vibration*, since 2023
- *Chaos: An Interdisciplinary Journal of Nonlinear Science*, since 2022
- *Mathematics*, since 2022
- *Electronics*, since 2022
- *Entropy*, since 2021
- *Applied Sciences*, since 2021
- *Journal of Computational and Nonlinear Dynamics*, since 2021
- *Nonlinear Dynamics, Chaos, and Complexity: In Memory of Valentin Afraimovich (1945–2018)*, edited by Dimitri Volchenkov, Springer, 2021
- *International Journal of Bifurcation and Chaos*, since 2020
- *International Journal of Dynamics and Control*, since 2020
- *ISA Transactions*, since 2020
- *Journal of Vibration Testing and System Dynamics*, since 2019
- *The European Physical Journal Special Topics*, since 2019
- *Journal of Applied Nonlinear Dynamics*, since 2016

➤ Referee for Conference Papers

- *International Design Engineering Technical Conferences & Computers and Information in Engineering Conference*, since 2019
- *International Mechanical Engineering Congress & Exposition*, since 2019

Professional Membership

Member American Society of Mechanical Engineers (ASME), since 2015